

# Dense Passage Retrieval for Open-Domain Question Answering

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**Abstract** — Open-domain question answering relies on efficient passage retrieval to select candidate contexts for answer extraction. Traditional sparse vector space models such as BM25 match keywords effectively but fail to capture latent semantic intent. In this work, we propose Dense Passage Retrieval (DPR), demonstrating that retrieval can be implemented using dense dual-encoder representations learned on a small set of question-passage pairs. When evaluated on Top-20 retrieval accuracy across Natural Questions (NQ) and TriviaQA, DPR outperforms BM25 by 9%-19% absolute gain. Our findings establish that dense embeddings can replace traditional sparse indices for knowledge-intensive NLP tasks.

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## 1. Introduction

Open-domain question answering (QA) is a benchmark task in natural language processing. Standard systems follow a two-stage retrieve-and-read paradigm. Historically, the retrieval step has been dominated by classical sparse models like TF-IDF and BM25. While computationally efficient and robust, BM25 exhibits fundamental lexical bottlenecks: it cannot resolve synonyms, paraphrases, or semantic context.

## 2. Methodology & Dual-Encoder Architecture

Our architecture uses two independent BERT-base encoders: a question encoder  $EQ(\cdot)$  that maps question text to a  $d$ -dimensional vector, and a passage encoder  $EP(\cdot)$  mapping passage text to the same space. Similarity is defined as the inner product  $\text{sim}(q, p) = EQ(q) \cdot EP(p)$ . We train the dual-encoder using in-batch negative sampling and cross-entropy loss against gold question-passage pairs. Passages are segmented into 100-word blocks and indexed using FAISS for sub-millisecond MIPS search.

## 3. Experimental Setup & Datasets

We benchmark DPR on five open-domain datasets: Natural Questions (NQ), TriviaQA, WebQuestions, CuratedTREC, and SQuAD. Evaluation metrics focus on Top-20 and Top-100 retrieval accuracy (the percentage of questions where at least one top- $k$  passage contains the ground-truth answer). For comparison, we implement an optimized Lucene BM25 baseline.

## 4. Results & Key Findings

On Natural Questions, DPR achieves 78.4% Top-20 retrieval accuracy compared to 59.1% for BM25 (a +19.3% absolute improvement). On TriviaQA, DPR reaches 79.4% Top-20 accuracy vs 75.4% for BM25. In reader downstream accuracy, DPR improves Exact Match (EM) scores by 4.2 points on NQ. The results demonstrate that dense semantic representations uniformly outperform sparse lexical matching in open-domain benchmarks.

## 5. Limitations & Discussion

Our approach is limited by domain generalization. On highly specialized technical or medical vocabularies not observed in training, pure DPR suffers from out-of-vocabulary degradation where BM25 remains robust. Furthermore, maintaining high-dimensional 768-d FAISS indices for 21M Wikipedia passages requires over 65GB of RAM, posing significant hardware overhead.

## 6. Conclusion

We introduced Dense Passage Retrieval (DPR), proving that dual-encoder dense representations are superior to sparse lexical matching for open-domain QA. Future work should investigate hybrid sparse-dense indices and index compression techniques.